Om Patel

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EDUCATION

Bachelor of Cybersecurity (Honours), Co-op Option September 2025–Present

Carleton University, Ottawa, Ontario

- Second year standing, Overall CGPA: 11.20/12.0 (A), Major CGPA: 11.00/12.0 (A)
- Expected Graduation Date: April 2030

AVAILABILITY

Available for 4 months beginning May 2027

RELEVANT SKILLS, EXPERIENCES AND ACCOMPLISHMENTS

Systems Programming

- Building a small C compiler: a lexer, a recursive-descent parser, and a little x86-64 assembly, assembled and checked against GCC
- Python and JavaScript for endpoint defense: a host daemon using psutil and watchdog, and a Chrome extension on Manifest V3
- Java and Python from coursework, including object-oriented design, data structures, and JUnit tests
- Bash, Git, GCC, and GDB for builds, version control, regression checks, and debugging

Security and Application Development

- Session-token defense that clears authentication cookies when a tab closes and stops unauthorized reads of the cookie database in under 500 ms on Windows and Linux
- Shipped EchoSpace, a keyboard-driven terminal mixer in Python and Textual, with a live NumPy spectrum, a timer, notes, and a task list
- Checked SessionSentinel with a local proof-of-concept reader on Windows and Linux, and the compiler with Bash scripts against GCC
- Coursework in C, Python, Java, JavaScript, HTML, and CSS. x86-64 assembly is limited to the small compiler, and I am not fluent in it

Communication & Collaborative Skills

- Taught programming and robotics fundamentals to over 50 high school students in virtual hackathons, turning complex algorithms into accessible coding exercises
- Authored detailed technical documentation and lecture synthesis notes for university computer science courses, delivering critical study resources to peers within strict 24-hour deadlines
- Communicated project requirements and technical feedback across diverse teams in English, Hindi, and Gujarati, bridging the gap between technical and non-technical audiences

APPLIED PROJECTS

Systems and Security Developer October 2026–Present

SessionSentinel (Complete) | JavaScript, Python, psutil, watchdog | Windows and Linux | github.com/Tz-risk/SessionSentinel

- Built a Manifest V3 Chrome extension that maps open tabs and purges that host's authentication cookies when the tab closes
- Wrote a Python daemon with watchdog and psutil that terminates unauthorized readers of Chrome's cookie database in under 500 ms
- Canonical path checks keep VS Code, Discord, and Spotify from being treated as stealers
- Validated the finished Windows and Linux builds with a local proof-of-concept reader, rogue_stealer.py

Lead Systems Developer May 2026–Present

Small C Compiler (In Progress) | C, x86-64, GCC, GDB, Bash

- Writing a basic compiler in C for a small subset of the language: a lexer, a recursive-descent parser, and an abstract syntax tree
- Emits a little x86-64 assembly, then assembles and links a binary. I am not fluent in assembly
- A Bash harness checks the binaries against GCC, and GDB is used while debugging
- Still in progress. The current subset is small, including return statements and integer constants

Software Developer October 2026–Present

EchoSpace (Shipped) | Python, Textual, NumPy | github.com/Tz-risk/EchoSpace

- Built a keyboard-driven terminal mixer that layers ambient tracks and draws a live spectrum with NumPy
- Keeps a timer, notes, and a short task list in the same session

WORK EXPERIENCE

Customer Service & Operations Team Member October 2022–March 2023

Tim Hortons, Edmonton, Alberta

- Managed high-volume order processing and drive-thru service under strict morning rush deadlines, multitasking between order taking and food preparation to maintain rapid drive-thru turnaround
- Reconciled end-of-shift inventory records and cash registers with zero discrepancies, collaborating closely with shift supervisors to maintain high workplace safety and quality control standards

VOLUNTEER & LEADERSHIP EXPERIENCE

Volunteer Academic Notetaker January 2026–Present

Carleton University (Ventus Student Accommodations), Ottawa, Ontario

- Summarized complex computer science and discrete math lectures into clean study documents, providing learning accommodations to peers within strict 24-hour deadlines
- Maintained strict confidentiality and reliable attendance throughout the term, earning a commendation from the Paul Menton Centre for thorough and consistent service

Coding Hackathon Workshop Host & Coordinator November 2023–June 2024

Peel District School Board (PDSB), Brampton, Ontario

- Organized virtual programming workshops for over 50 high school students, running live coding tutorials on VEX platforms to teach basic algorithms and control flow
- Partnered with teachers to monitor breakout rooms and fix software issues in real time, helping students finish their projects and build their interest in STEM